



Machine Learning and Deep Learning Methods for Electricity Day-Ahead Price Forecasting

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Agenda



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Introduction

- i. Motivation
- ii. Literature Review
- iii. Data Overview

Motivation



- ❑ Future **power production is uncertain** due to:
 - ❑ **Environmental factors**
 - ❑ **Energy source variability**
- ❑ Knowing electricity costs beforehand helps:
 - ❑ **Anticipate shortages** to avoid blackouts
 - ❑ Identify locations for infrastructural investment
- ❑ Accurate **electricity price forecasts are essential to maintain social and economic stability**

Literature Review

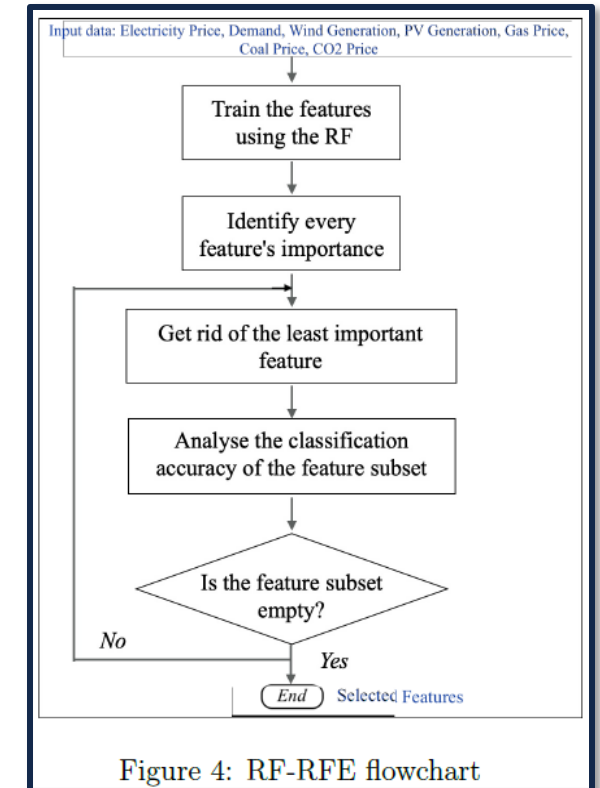


❑ Traditional Approaches to Electricity Price Forecasting:

- ❑ Short-term electricity price forecasting (EPF) has traditionally relied on **statistical and econometric models** (e.g., ARX, LASSO, regime-switching) frameworks
- ❑ Machine learning and deep learning models improve performance by capturing **nonlinear and volatile price dynamics**

❑ Feature Selection:

- ❑ RF-RFE ranks feature importance using Random Forest and iteratively removes the least important features to select an informative subset



Based on: Ben Amor, S., Möbius, T., & Müsgens, F. (2024). *Bridging an energy system model with an ensemble deep-learning approach for electricity price forecasting.*

Literature Review



❑ Limitations:

- ❑ Purely data-driven models do not explicitly capture physical and economic market mechanisms
- ❑ While hybrid approaches improve forecasting accuracy, they rely on complex energy system models and increased computational cost → limits **scalability** and **practical deployment**

❑ Positioning of Our Project:

- ❑ Aligns closely with the literature by adopting the same market (German EPEX SPOT), forecasting horizon (hourly day-ahead), and core price drivers
- ❑ Acknowledges the limitations of purely data-driven models, as highlighted in recent hybrid electricity price forecasting studies
- ❑ Focus on practical, scalable **deep learning** and **boosting models**



Data Overview

Data Overview

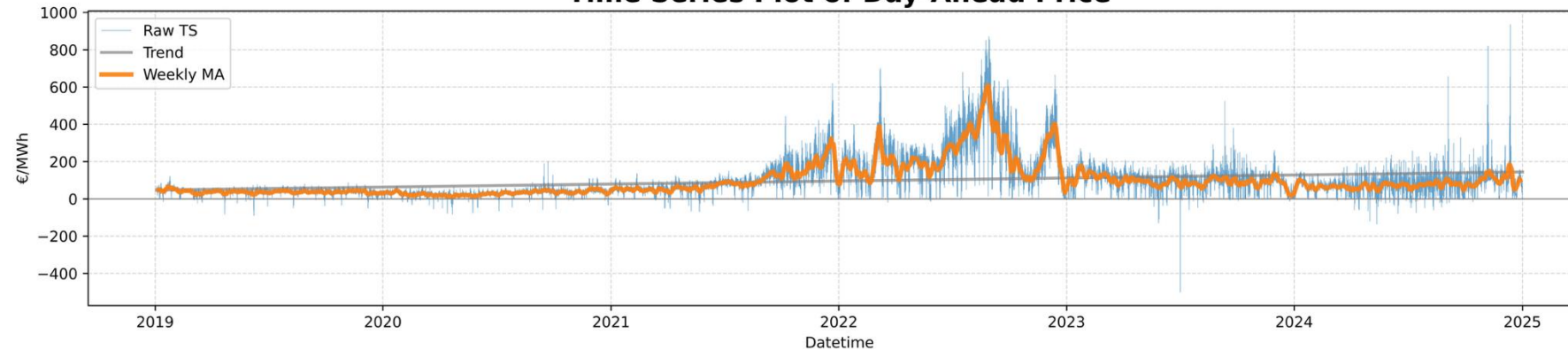


- ❑ **Time Range:** Hourly electricity prices from January 1, 2019 to December 31, 2024
- ❑ **Target Variable:**
 - ❑ Day-Ahead Electricity Price (EUR/MWH) from the German EPEX SPOT market
- ❑ **Raw Data Sources:**
 - ❑ Bright Sky API → Weather data
 - ❑ Energy Charts API → power system and price data
- ❑ **Collected Variables:**
 - ❑ Weather: precipitation, cloud cover, temperature, relative humidity
 - ❑ Power System:
 - ❑ Load: actual grid load (MW), recorded at the end of the following hour
 - ❑ CBET: total cross-border electricity trading (MW)
 - ❑ Market Price: day-ahead hourly electricity price (target variable)

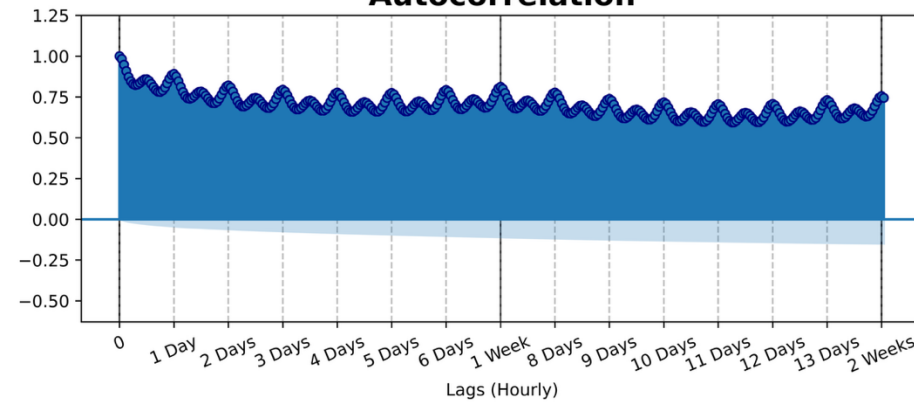
Data Overview



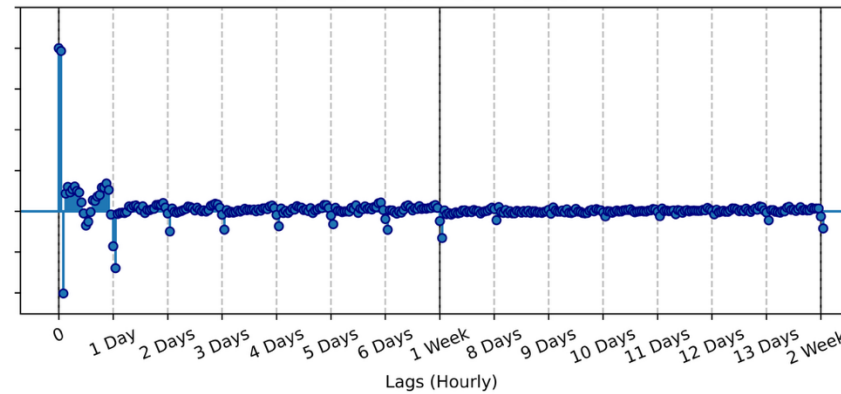
Time Series Plot of Day-Ahead Price



Autocorrelation



Partial Autocorrelation



- ☐ Day-ahead price is **stationary**
- ☐ **Clear periods of high volatility**

Autocorrelation Function Plot (ACF):

- ☐ Daily (24-hour) seasonality
- ☐ Potential weekly (168-hour) seasonality

Partial Autocorrelation Function (PACF):

- ☐ Daily seasonality diminishes after ~8 days
- ☐ Weekly seasonality diminishes after ~2 weeks
- ☐ Last day is most correlated
- ☐ Last week is next most correlated



Approach

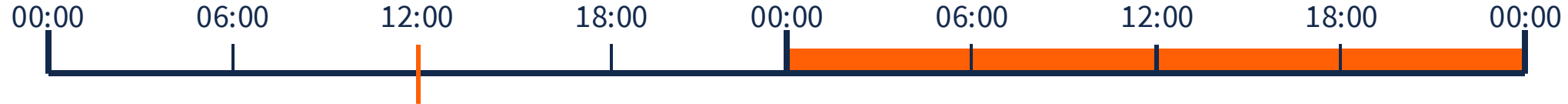
- i. Preprocessing
- ii. Modeling



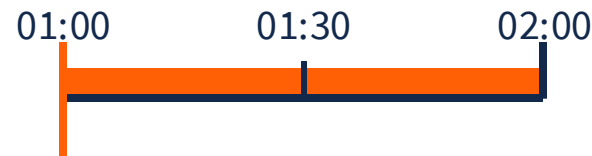
Preprocessing

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Feature Engineering

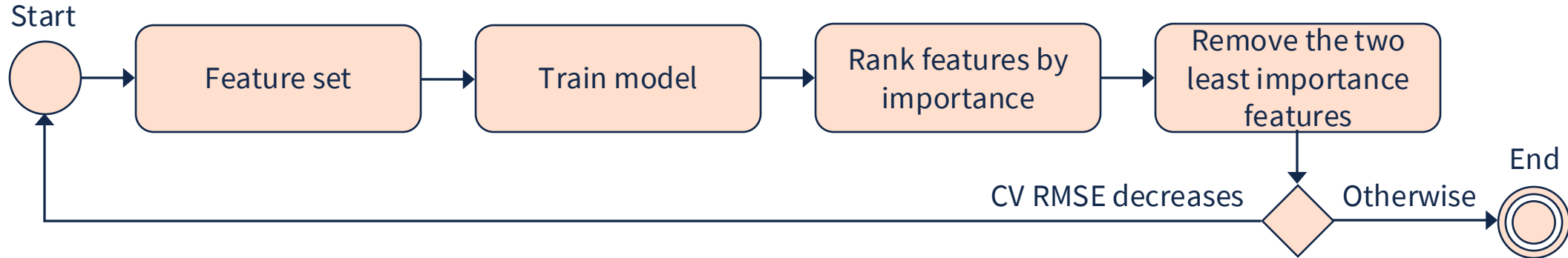


- ❑ Traders gather at noon of the current day to determine hourly prices for all the next day
- ❑ Accordingly, all **covariates were lagged between 24 – 48 hours** ($l \in \{24, 25, \dots, 48\}$)
- ❑ This avoids overcomplicated solutions and **incorporates real-time forecasting constraints**



- ❑ Additionally, all **variables were carefully aligned** according to research / industry standards
 - ❑ Each timestamp represents the data for the following hour
 - ❑ e.g., *(01:00) cumulative rainfall until 02:00*
- ❑ **Lagged price** was included as a covariate in the dataset

Recursive Feature Elimination (RFE)



- ❑ **Dimensionality reduction** technique
- ❑ Cycle recursively eliminates features based on internal model **feature importance rankings**
- ❑ **Light Gradient Boosting Machine** (LGBM) with default hyperparameters was used for ranking
- ❑ **Cross-validation** (CV) was used for stopping
 - ❑ 2 splits
 - ❑ 90% train, 10% test

Target Transformations



Min-Max Scaling

Transformation

$$y_{\text{norm}} = \frac{y - \min(y)}{\max(y) - \min(y)},$$

i. e., $y_{\text{norm},i} \in [0,1]$ for all i

Inverse
Transformation

$$y = y_{\text{norm}} \cdot (\max(y) - \min(y)) + \min(y)$$

Effect

- ☐ Accelerates **model convergence**
- ☐ **Reduces effect of spikes** in price on fit
- ☐ Improves model **generalizability**

Mirror-Logarithm

$$y_{\text{mlog}} = \text{sgn}(y) \left[\log \left(|y| + \frac{1}{c} \right) + \log(c) \right],$$

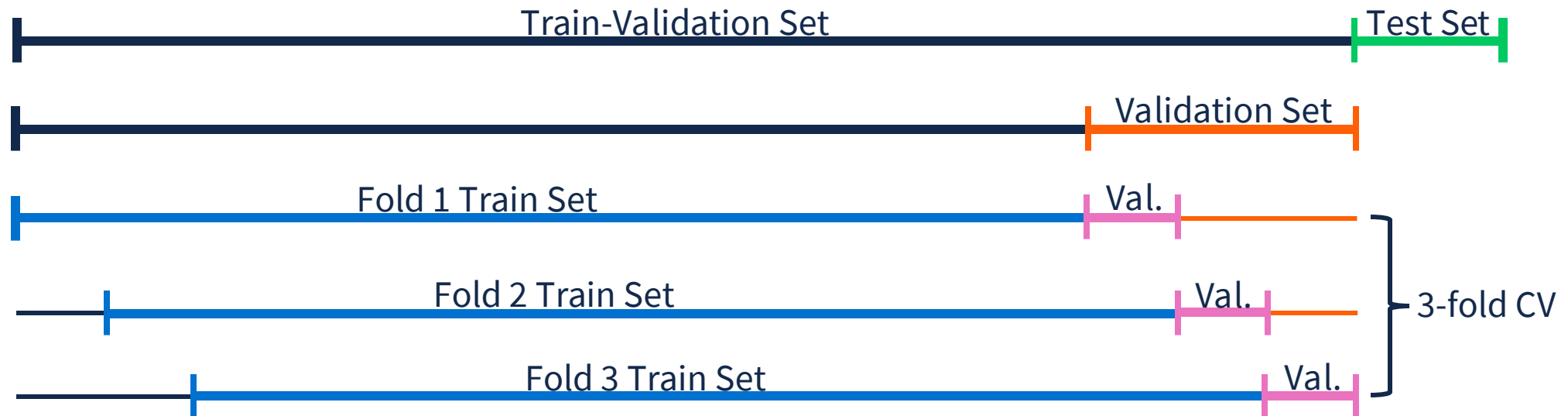
i.e., $c = 1/3$

$$y = \text{sgn}(y_{\text{mlog}}) \left[e^{|y_{\text{mlog}}| - \log c} - \frac{1}{c} \right],$$

i.e., $c = 1/3$

- ☐ **Generalization of logarithm** for zero and negative values
- ☐ Log-damping effect
- ☐ Accelerates **model convergence**
- ☐ Proven to **reduce effect of spikes** in price on fit

Train-Validation-Test Split



- ☐ **72% - 18% - 10%** split (train - validation - test)
- ☐ **3-fold temporal CV** for hyperparameter optimization
- ☐ **Rolling window** method:
 - ☐ Preserves **time order**
 - ☐ **Prevents data leakage** during CV



Modeling

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Machine Learning Models



Support Vector Regression (SVR)

Gradient Boosting Regression

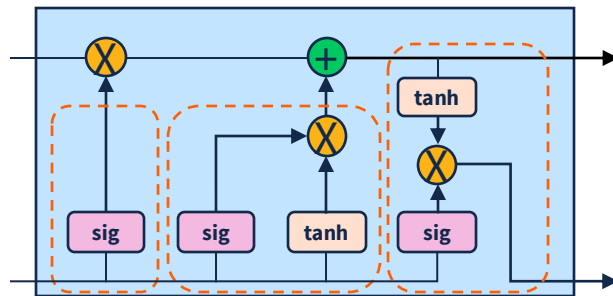
Model Formula	$\hat{y}_i = \sum_{n=0}^N (\alpha_n - \alpha_n^*) K(x_n, x) + b,$ i. e., $K(x_n, x) = \exp(-\gamma x_n - x ^2)$	$\hat{y}_i = \sum_{k=1}^t f_k(x_i)$
Key Hyper-parameters	$\gamma \in (0, \infty)$: model complexity $C \in (0, \infty)$: regularization	$t \in \mathbb{Z}_{\geq 1}$: number of trees depth $\in \mathbb{Z}_{\geq 1}$: maximum depth of a tree learning rate $\in (0, \infty)$: step size shrinkage
Notes	❑ Radial basis function (RBF) kernel handles nonlinear dependencies ❑ Robust to outliers ❑ Built-in regularization	❑ Extreme (XGBoost), Light (LGBM), Categorical (CatBoost) ❑ Sequential learning from boosting ❑ Handles nonlinear dependencies ❑ Built-in regularization ❑ Scales better for large datasets

Deep Learning Models



Long Short-Term Memory (LSTM) Model

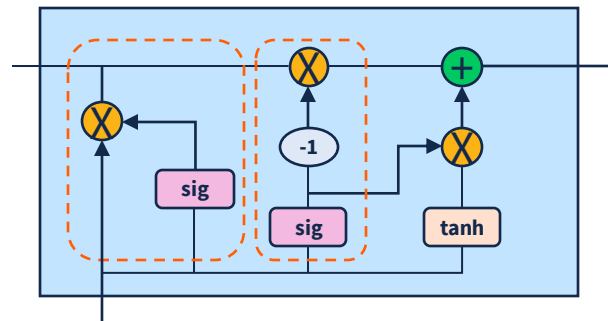
- ❑ 3 gates per cell: input, forget, output
- ❑ Good for **long temporal dependencies**
- ❑ Captures nonlinear relationships



Simplified process within an LSTM cell

Gated Recurrent Unit (GRU) Model

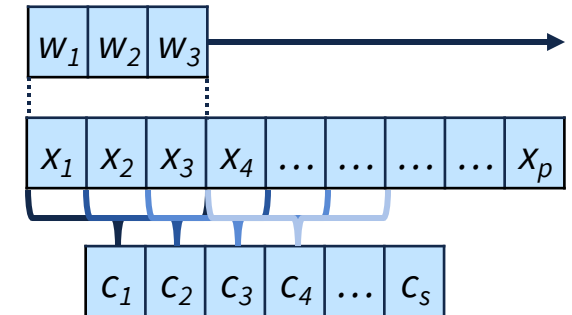
- ❑ 2 gates per cell: update, reset
- ❑ Simplified LSTM
- ❑ **Faster training** & fewer parameters
- ❑ Good for **moderately-long temporal dependencies**



Simplified process within a GRU cell

Convolutional Neural Network (CNN) + LSTM Model

- ❑ Compress / **extract local patterns** before LSTM
- ❑ Capture **short-term temporal dependencies** (CNN) alongside long-term ones (LSTM)



Simplified process within a 1D CNN layer

Hyperparameter Tuning



Grid Search CV

Random Search CV

Bayesian Optimization (BO) CV + Early Stopping

Notes

- ☐ **Inefficient** for large, high dimensional search spaces
- ☐ **Iteratively develop accurate search ranges for future BO** on SVR

- ☐ **Inefficient** for large, high-dimensional search spaces
- ☐ **Iteratively develop accurate search ranges for future BO** on boosted models

Tree-structured Parzen Estimator (TPE) as sampler

- ☐ Supports many types of spaces
- ☐ Non-parametric & directly models good / bad regions

Hyperband as pruner

- ☐ Reduces runtime drastically
- ☐ Reduces wasteful trials

Models
Tuned

SVR

XGBoost, LGBM, CatBoost

SVR, XGBoost, LGBM, CatBoost,
LSTM, GRU, CNN-LSTM



Findings

- i. Reduced Dataset
- ii. Effect of Target Transformations
- iii. Model Evaluations

Reduced Dataset



Covariate	Lags (Hourly)	Units
Price	24, 31, 37, 41, 43, 45, 47, 48	€ / MWh
Temperature	24, 28, 33, 40, 46	°C
Humidity	29, 36, 48	%
Load	24, 28, 30, 34, 36, 42, 48	MW
CBET	24, 28, 32, 36, 43, 48	GW
29 total covariates		

Effect of Target Transformations



Min-Max Scaling

- ☐ **Reduces effect of high variances** price periods on model fit
- ☐ **Faster model convergence** (especially for complex neural network models)
- ☐ Stable gradients during neural network training

Mirror-Logarithm

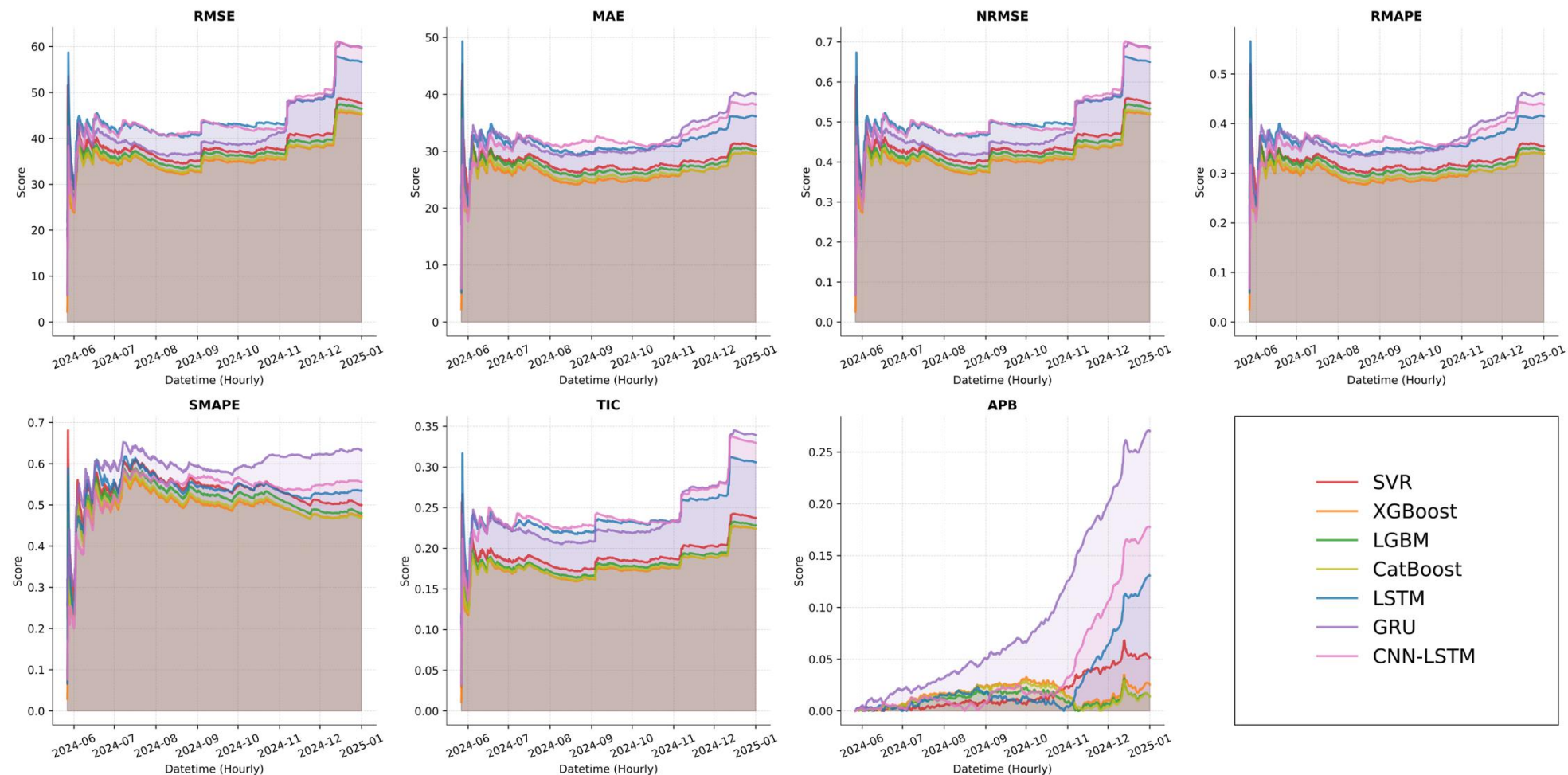
- ☐ **Explosion of several point transformations**
- ☐ Unsure origin, but likely due to floating point error
- ☐ Rendered unusable for modeling

Min-Max scaling was chosen as the target transformation method



Model Evaluations

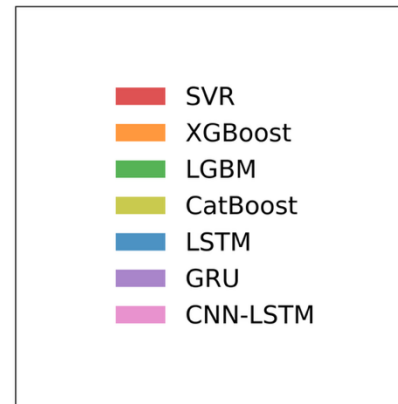
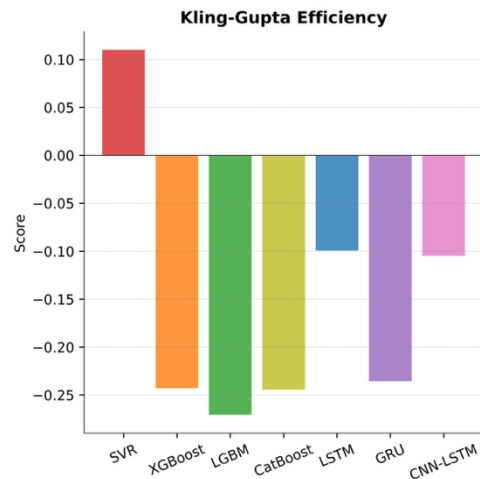
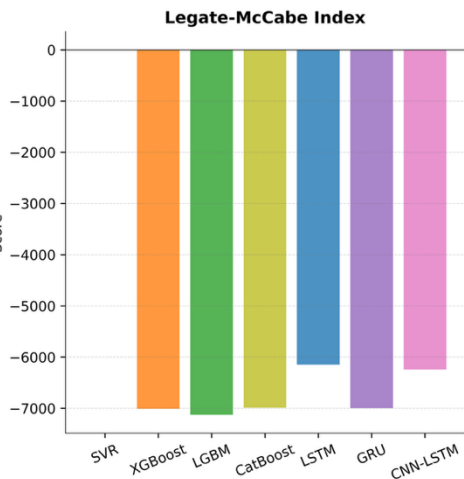
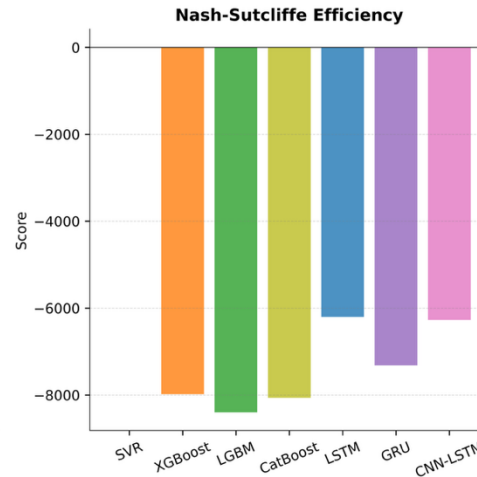
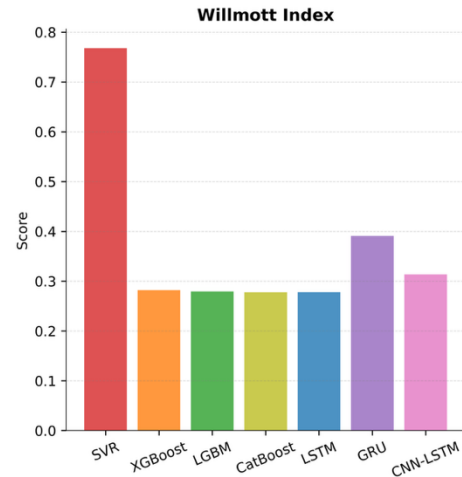
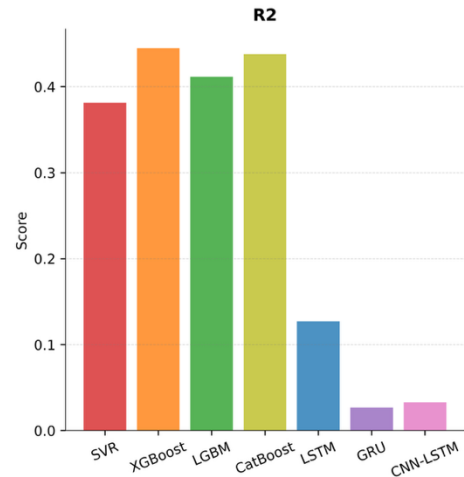
Bias Evaluations



Accumulative bias metrics on test set:

- ❑ Descending order of average bias: ANNs, SVR, boosting
- ❑ Similar metric scores among boosted models, with **XGBoost often showing best performance**
- ❑ **Sections of high price volatility** causing spikes in bias over time across all models
- ❑ **Volatility spikes are higher for ANNs**

Variance Evaluations



Overall variance metrics on test set:

- ☐ **XGBoost has best R²**, although comparable to other boosted models
- ☐ **SVR** accounts for most variance in **all metrics other than R²** by far
- ☐ Both **SVR and boosted models** tend to **follow inherent variance structure** moderately well in the test set
- ☐ **ANN models** capture least amount of variance structure in the test set

Deep learning models tend to underestimate spikes in price, causing them to lose predictive power during volatile periods.



Discussion

- i. Strengths
- ii. Limitations
- iii. Future Work

Strengths



☐ **Well-Designed Data Pipeline**

- ☐ Applied consistent preprocessing, including mirror-log transformation and Min–Max normalization, reducing volatility and stabilize model training.

☐ **Structured Feature Engineering**

- ☐ Applied systematic feature selection using LGBM to eliminate redundant inputs and streamline the modeling dataset.
- ☐ Verified that all engineered features adhered to the day-ahead auction timing (no look-ahead leakage).

☐ **Robust and Fair Model Comparison**

- ☐ Used rolling-window evaluation to simulate real forecasting conditions and reduce overfitting.
- ☐ Evaluated seven distinct modeling approaches under the same preprocessing and validation framework.

Limitations



- ❑ **Limited Covariates:**

- ❑ Dataset includes only weather variables, grid load, and cross-border flows (CBET)

- ❑ **Modeling Limitations:**

- ❑ Limited hyperparameter optimization for deep networks due to time and compute constraints

- ❑ **Feature Engineering Limitations:**

- ❑ Lag features capture short term patterns but miss long-range structure (lack of weekly, monthly, yearly seasonality)

- ❑ **Evaluation & Validation Limitations:**

- ❑ Rolling validation depth is limited, validation folds do not span all major structural changes (COVID, energy crisis, Russia-Ukraine conflict)
 - ❑ Lack of regime-specific evaluation, models are not separately evaluated under normal, volatile, and extreme-price conditions

Future Work



☐ **Broaden Covariates:**

- ☐ Add fuel prices, CO₂ prices, average solar radiation, and wind speed
- ☐ Include market sentiment or macroeconomic indicators that affect energy demand

☐ **Adopt More Advanced Temporal Models:**

- ☐ Explore Transformer-Based Architectures (e.g., Informer, Temporal Fusion transformer)
- ☐ Explore multi-resolution models combining hourly, daily, weekly and seasonal patterns

☐ **Enhance Data Transformations:**

- ☐ Use dynamic scaling windows (e.g., rolling normalization) to reduce distribution shifts
- ☐ Model price differences or returns rather than absolute prices to reduce non-stationary

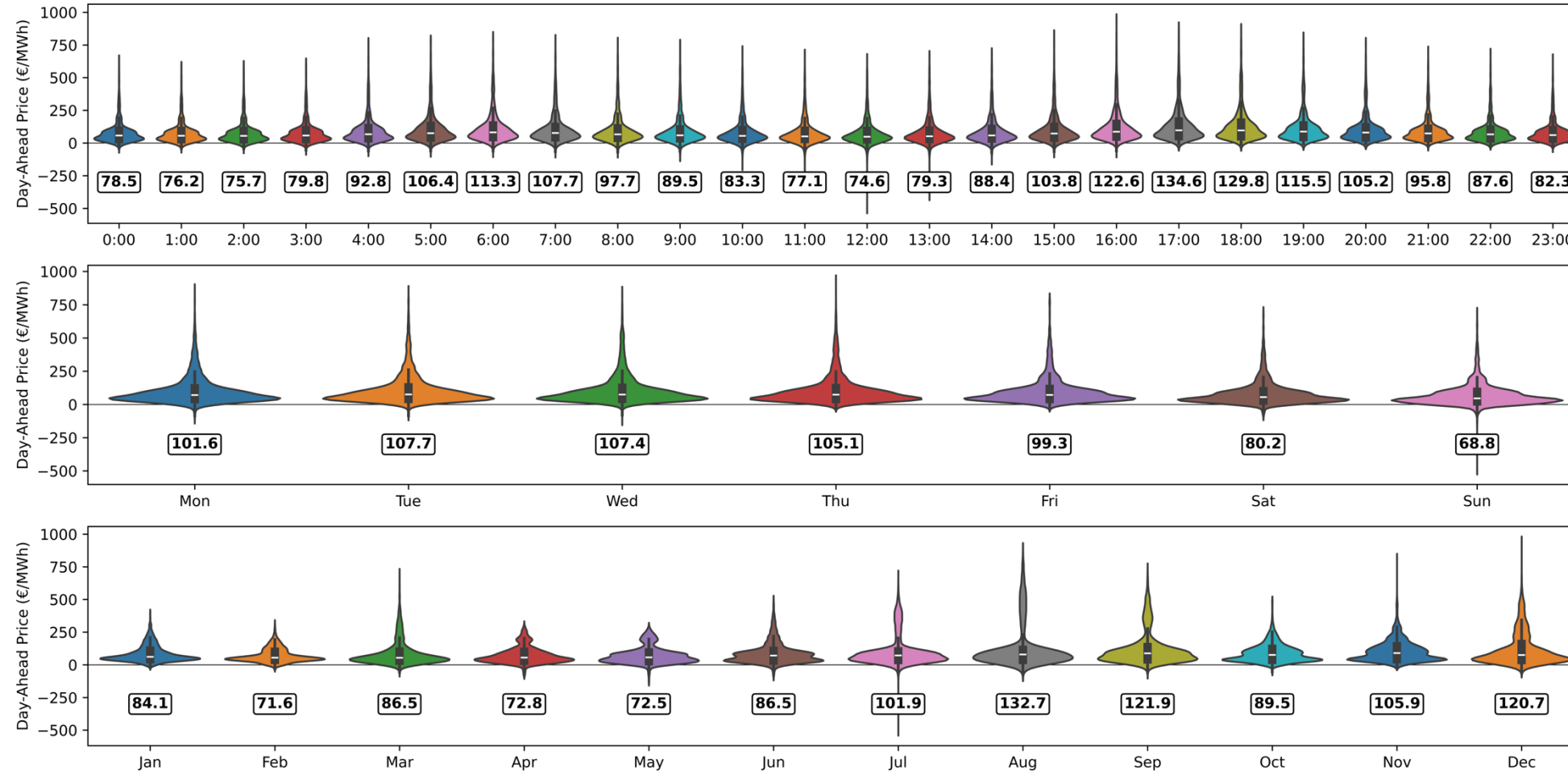
☐ **Incorporate Signal Processing:**

- ☐ Include signal decomposition techniques (e.g., Empirical Mode Decomposition, Variational Mode Decomposition) into the data processing pipeline
- ☐ Forecast decomposed signals separately and recombine to produce the final forecast



Appendix

Data Overview



Hourly:

- ☐ No strong differences
- ☐ Slightly higher prices in evening time

Daily:

- ☐ Little to no difference in price distribution

Monthly:

- ☐ Small differences in distribution by month
- ☐ Greatest variability in Summer & Winter



Thank You!

Questions?

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